

CCST9012 Tutorial Week 4

Observing Venus, Observing Mars

1 OBSERVING VENUS

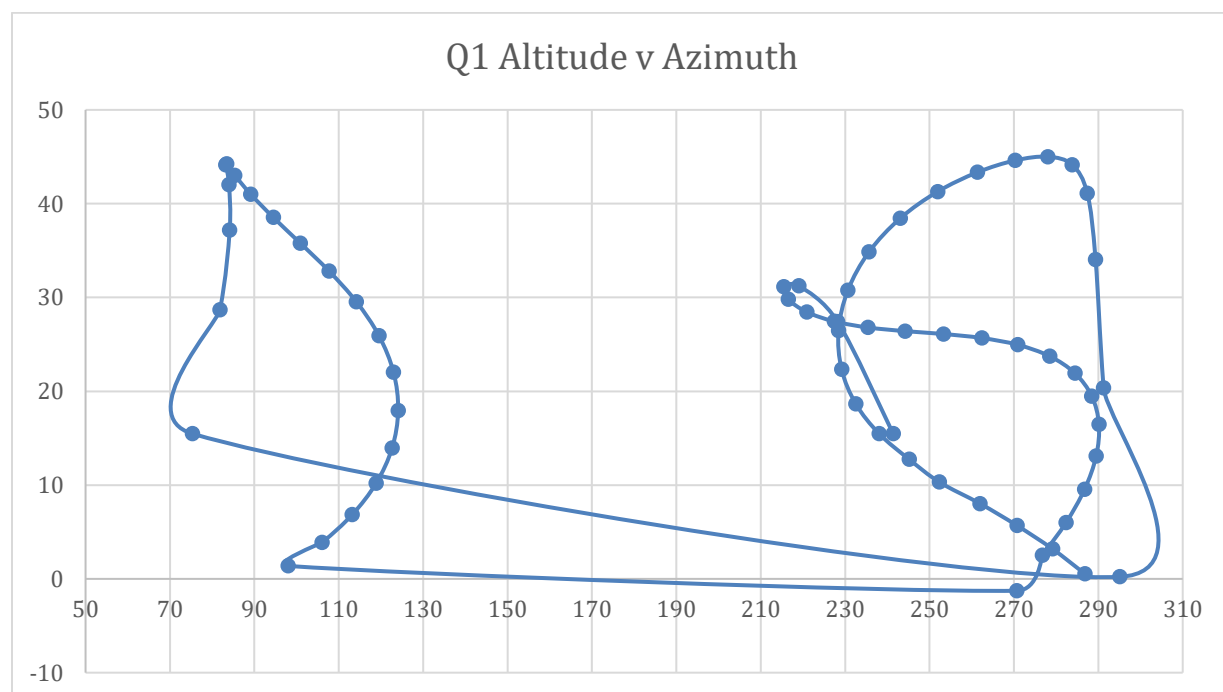
When visible, the planet Venus is very prominent in the night sky. Of particular interest to early astronomers was the fact that Venus (and Mercury) always appear close to the sun. In the lingo of this course, we say that the Inferior Planets are characterized by limited elongations.

This has several practical consequences:

- Looking in the direction of Venus is similar to looking in the direction of the sun and vice versa. It is impossible to stare at Venus while the sun is behind you.
- Because Venus is close to the sun, this means that it is usually above the horizon during daylight hours and our view of it is obscured by daylight. However, we can usually observe Venus either in the early morning hours before sunrise or the early evening hours after sunset.

In the spreadsheet file, *Venus Observations*, the data shows observations of Venus taken every 14 days from 14 August 2019 until 29 December 2021. The position of Venus in the Sky is observed either at sunrise or sunset.

Q1. Create a graph using this data that shows the altitude of Venus on the vertical axis and the azimuth on the horizontal axis.

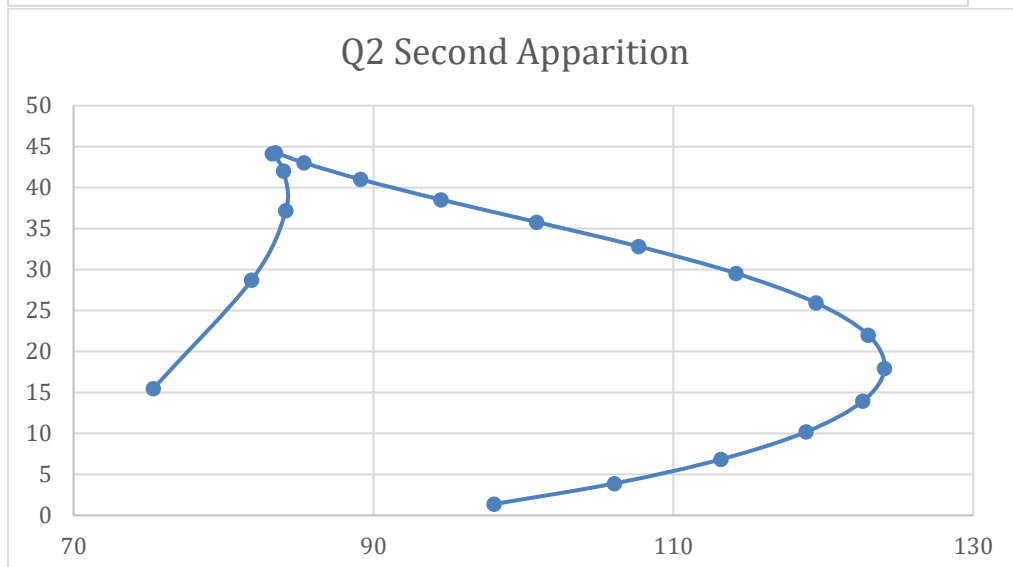
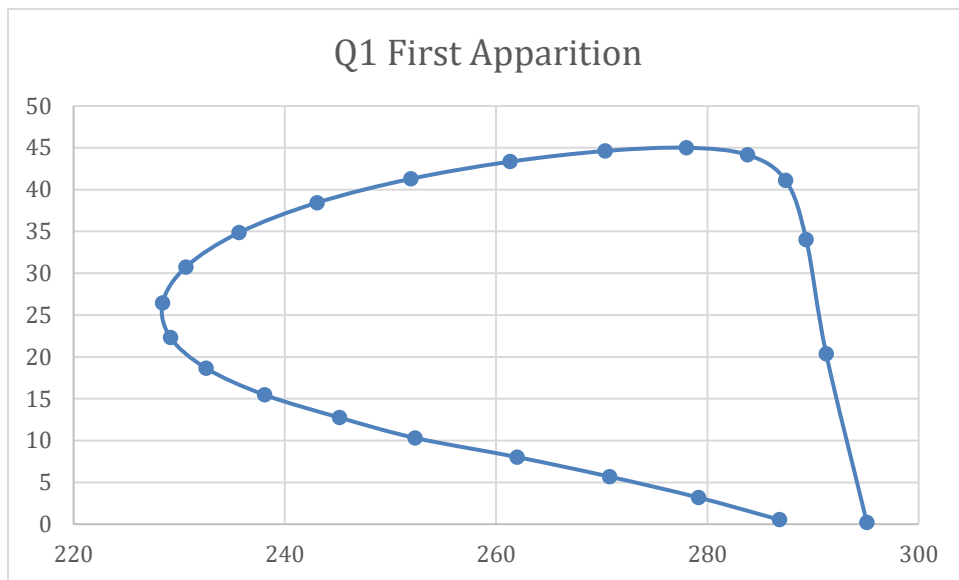


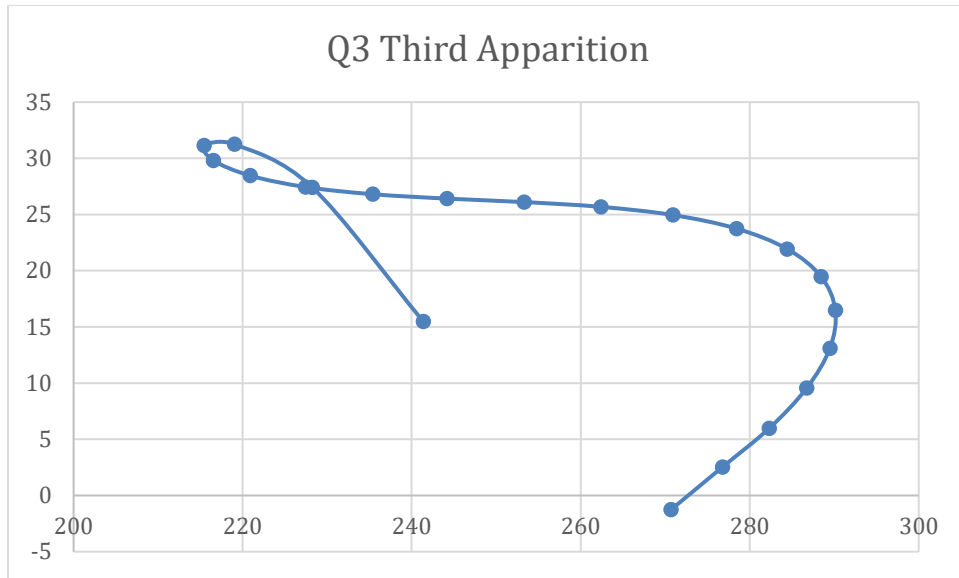
From the time that Venus appears as a morning or evening star until it disappears again is called an *apparition*. Each of these has a distinct shape or pattern. In an 8 year cycle, Venus will appear 5 times as a morning star and 5 times as an evening star, each with a unique pattern.

Q2. How many *apparitions* of Venus are present in the dataset?

There are 3 apparitions in the dataset.

- 11 September 2019 – 3 June 2020
- 17 June 2020 – 10 March 2021
- 24 March 2021 – 29 December 2021



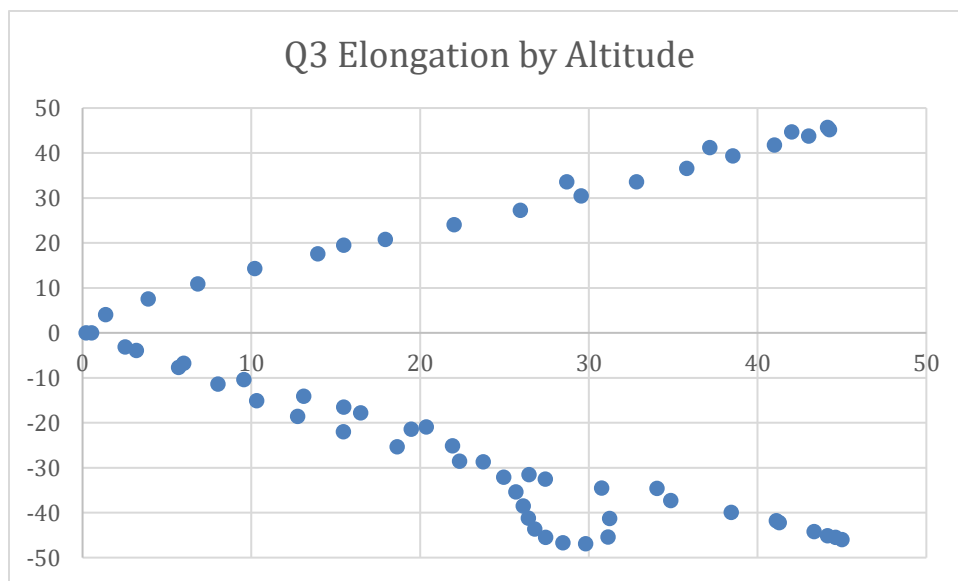


We have already revised that elongation refers to the angular separation between a planet and the sun. *Conjunction* refers to an elongation that is very close to zero, while *opposition* refers to elongation of 180° .

Because observations are made either at sunrise or sunset, the altitude of the Sun during each observation is the same.

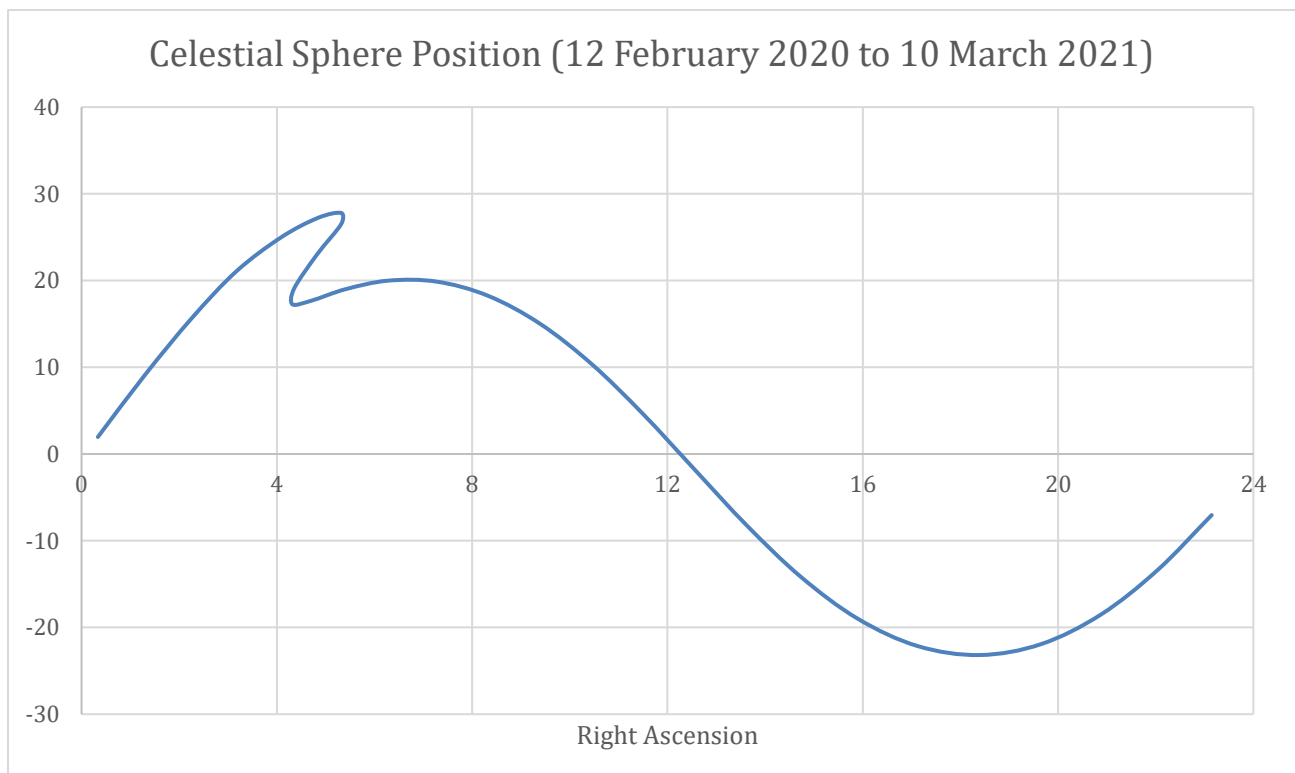
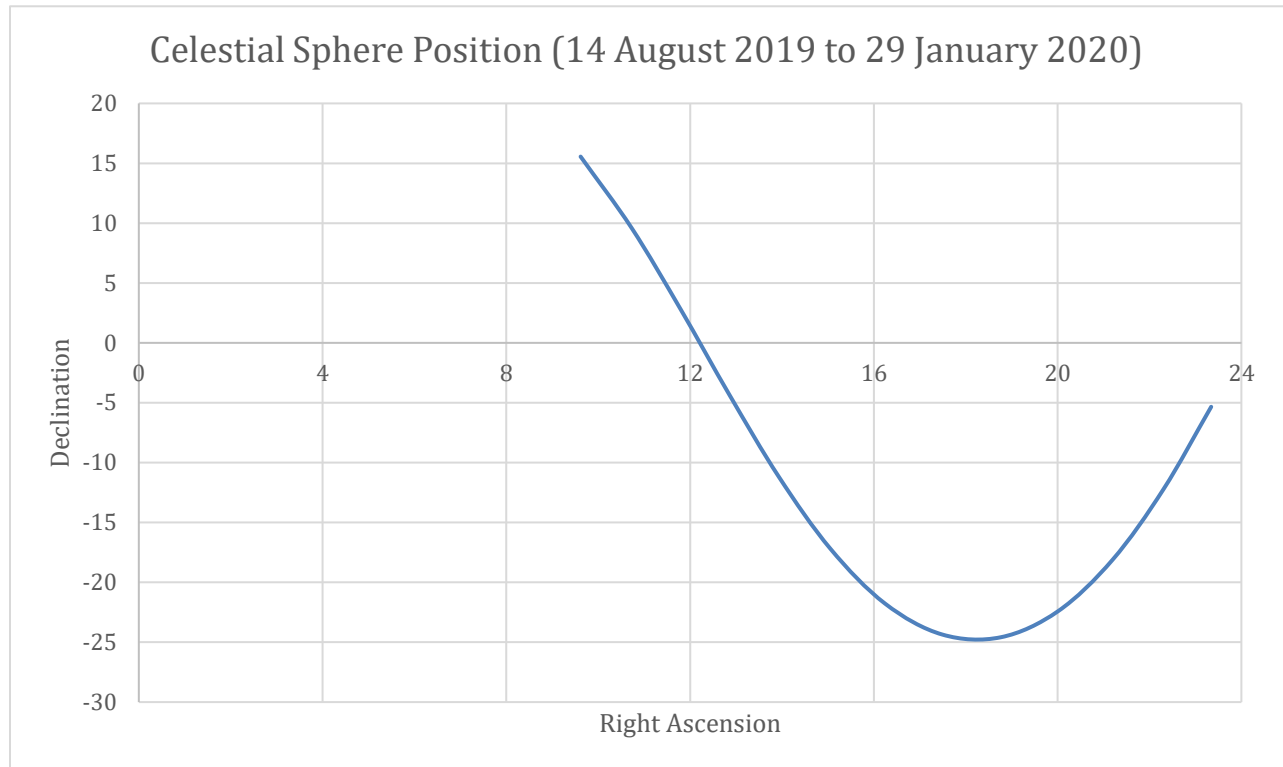
Q3. Show that the altitude of Venus gives an approximation of its elongation by creating a graph that shows the Elongation of Venus on the vertical axis and the Altitude of Venus on the horizontal axis.

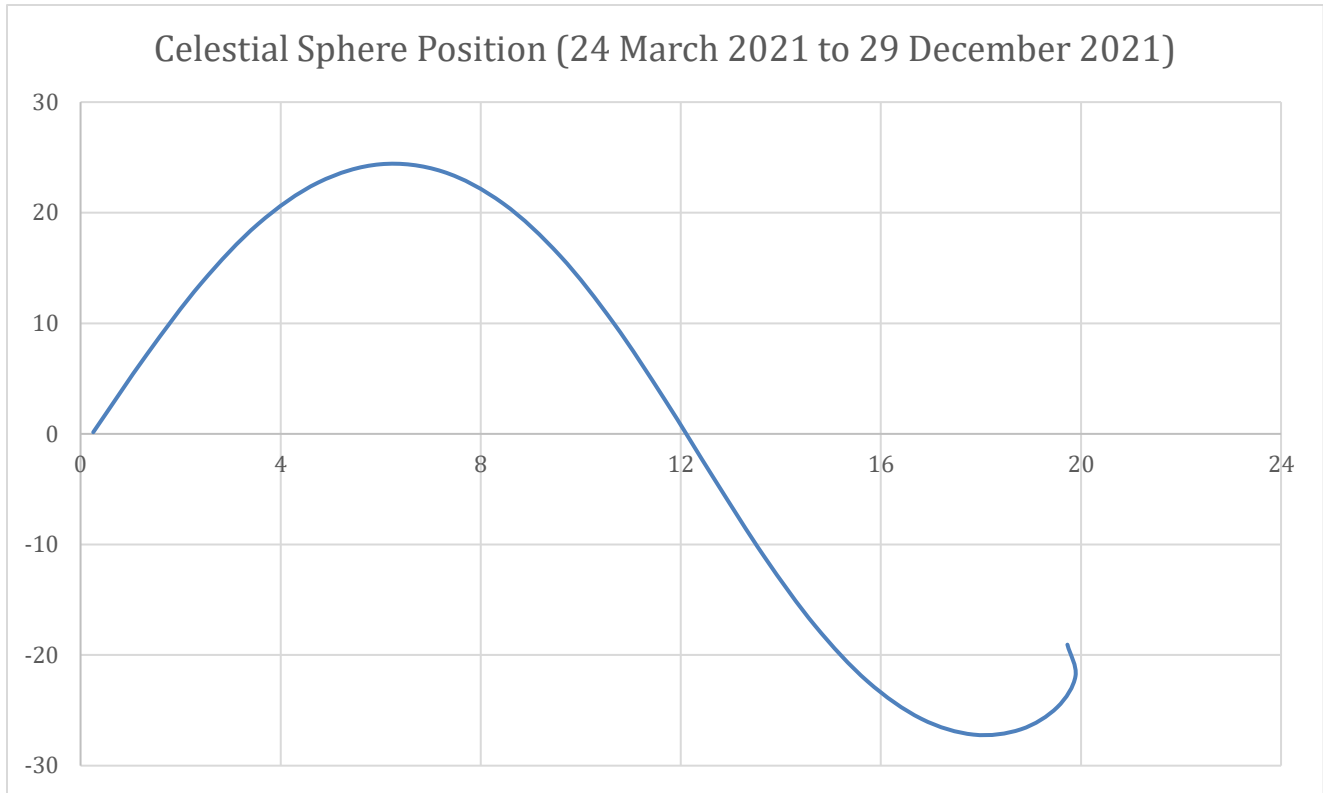
For the most part, the altitude is similar to the elongation. The extent that it differs is due to the angle that the ecliptic makes with the horizon which changes throughout a year.



Q4. Which data most directly indicate periods when Venus is in prograde and retrograde motion? During which dates is Venus in retrograde motion?

The most direct indication is the Right Ascension. When it is increasing, Venus is traveling from West to East on the Celestial Sphere and is in Prograde Motion. When it is decreasing, it is in Retrograde Motion.



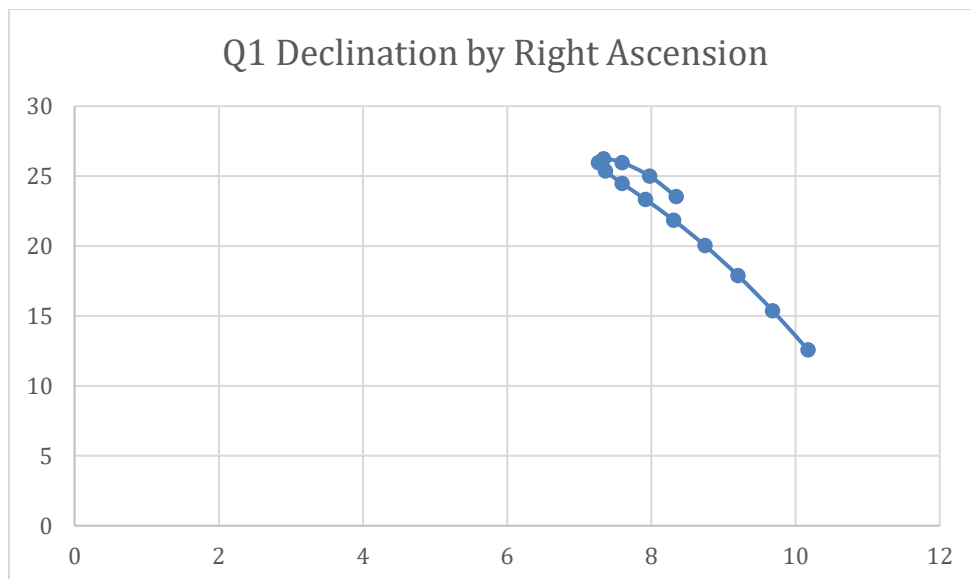
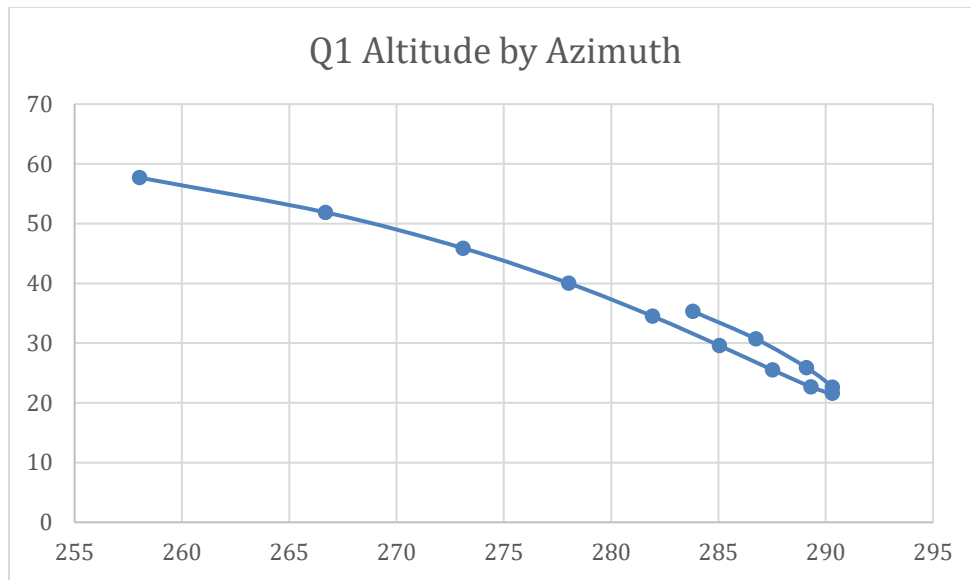


2 OBSERVING MARS

In contrast to Venus and Mercury, the superior planets may appear with any elongation to the sun. At times they are in opposition to the Sun, meaning that they pass through the zodiac constellation opposite of the Sun. Early astronomers were fascinated that at times of opposition, superior planets undergo their retrograde motion.

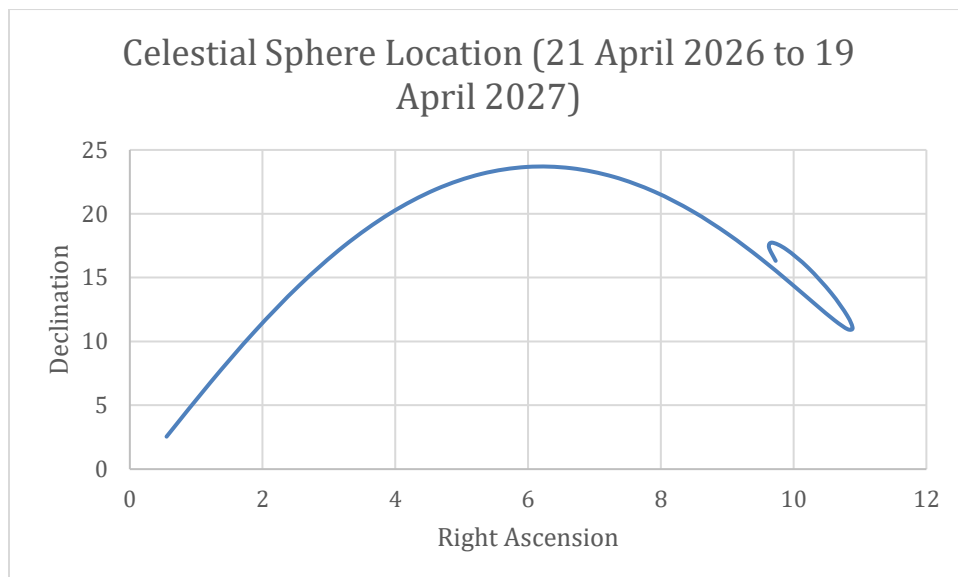
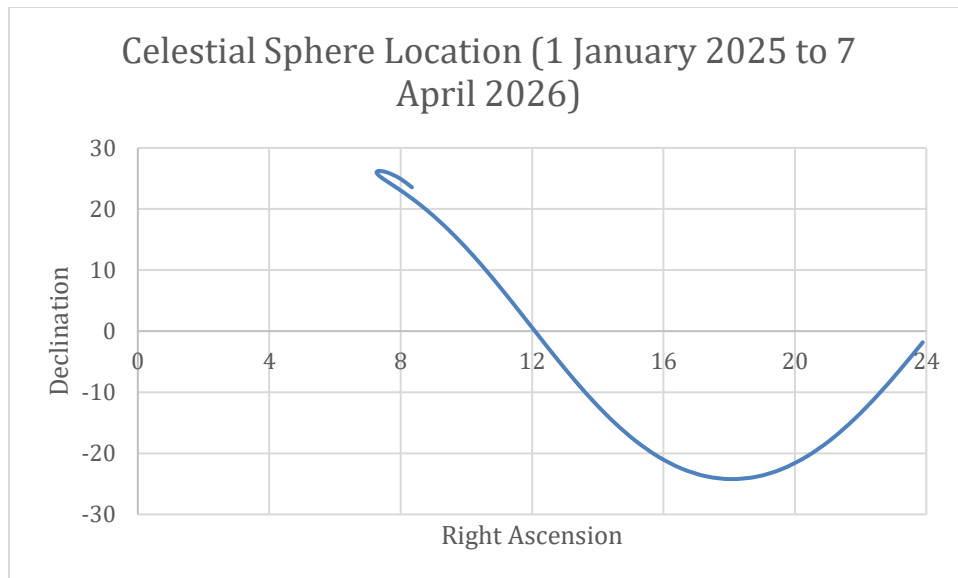
From 1 January 2025 to 17 June 2025, observations are separated by 14 sidereal days. Because the position of stars in the sky are identical when separated by sidereal days, the changing of azimuth and altitude indicate the movement of Mars relative to the fixed stars.

Q1. Create two graphs for this time period (1 January 2025 to 17 June 2025). In the first graph, show the altitude of Mars on the vertical axis and the azimuth on the horizontal axis. In the second graph, show the declination of Mars on the vertical axis and right ascension on the horizontal axis. Note similarities and differences between the two graphs.



In these two graphs show the same information at different scale. Note that increasing RA (West to East) is the opposite direction of movement as increasing Azimuth.

Q2. Observe the motion of Mars along the celestial sphere by creating a graph that shows the declination on the vertical axis and the right ascension on the horizontal axis.

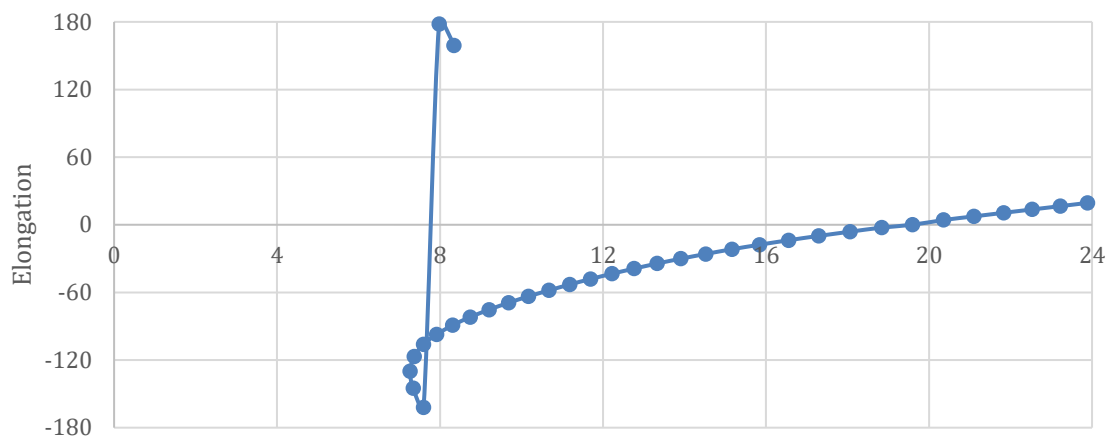


Q3. Identify dates in the dataset where Mars is in retrograde motion. Use these dates to estimate the *synodic* period of Mars. Verify the characteristic relationship with the Sun by creating a graph that showing the elongation on one axis and the right ascension on the other.

Retrograde Motion corresponds to observations where the Right Ascension is decreasing. In this dataset, Mars is in retrograde motion in the first observation, 1 January 2025 until 26 February 2025. Then it returns to prograde motion until the Right Ascension begins decreasing again 12 January 2027 until 5 April 2027.

The synodic period can be estimated as the time between 26 February 2026 and 5 April 2027 which is 768 days. Since our observations are typically 14 days apart. This is within acceptable error with the known value of 780 days.

Q3 Elongation by Right Ascension (1 January 2025 to 7 April 2026)



Q3 Elongation by Right Ascension (21 April 2026 to 19 April 2027)

